

C Programming Project

Topic : Password Strength Analyser

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Introduction :

The program analyzes the password entered by the user by checking for the presence of lowercase letters, uppercase letters, numbers, and special characters, which are essential criteria for creating a secure password. It uses a pointer to traverse each character of the password and evaluate its strength efficiently. Based on the number of conditions satisfied, the password is classified as weak, medium, or strong. If the password is found to be weak or medium, the program prompts the user to update it and repeats the verification process until a strong password is created or the user chooses to exit. By enforcing these rules, the program encourages better password practices and helps improve overall account security.

Explanation :

This program is used to check the strength of a password entered by the user. It repeatedly asks the user to enter a password and analyzes it

using a pointer to examine each character one by one. Four flag variables are used to check whether the password contains at least one lowercase letter, one uppercase letter, one digit, and one special character. The pointer starts at the beginning of the password string and moves through each character until the null character is reached, setting the corresponding flags based on the character type. After scanning the entire password, the program calculates the strength by counting how many of these conditions are satisfied. If all four conditions are met, the password is classified as STRONG, the message “It is a stronger one” is displayed, and the program ends. If only two or three conditions are satisfied, the password is classified as MEDIUM, and if zero or one condition is satisfied, it is classified as WEAK. In the case of a weak or medium password, the program asks the user whether they want to change the password and repeats the process until a strong

password is entered or the user chooses to exit the program.

Program :

```
#include <stdio.h>

#include <string.h>

int main()
{
    char password[100];
    char *ptr;
    char choice;
    while (1)
    {
        int lower = 0, upper = 0, digit = 0, special = 0;
        int count;
        printf("Enter password: ");
        scanf("%s", password);
        ptr = password;
        while (*ptr != '\0')
        {
```

```
    if (*ptr >= 'a' && *ptr <= 'z')
        lower = 1;
    else if (*ptr >= 'A' && *ptr <= 'Z')
        upper = 1;
    else if (*ptr >= '0' && *ptr <= '9')
        digit = 1;
    else
        special = 1;
    ptr++;
}
count = lower + upper + digit + special;
if (count == 4)
{
    printf("Password Strength: STRONG\n");
    printf("It is a stronger one\n");
    break;
}
else if (count >= 2)
```

```
{  
    printf("Password Strength: MEDIUM\n");  
}  
else  
{  
    printf("Password Strength: WEAK\n");  
}  
    printf("\nDo you want to change the  
password? (y/n): ");  
    scanf(" %c", &choice);  
    if (choice == 'n' || choice == 'N') {  
        printf(".....Program  
ended.....\n");  
        break;  
    }  
}  
return 0;  
}
```

Output :

Enter password: abc

Password Strength: WEAK

Do you want to change the password? (y/n):

y

Enter password: Abc123

Password Strength: MEDIUM

Do you want to change the password? (y/n):

y

Enter password: Abc@123

Password Strength: STRONG

It is a stronger one